

Full PI-VSG + CC-ICL — Nonlinear DAE Model

Matas-Díaz et al., JMPSCCE 2025, §II.A-II.C.2, eqs. (1)–(20)

System architecture

$\mathbf{dx/dt} = \mathbf{f}(\mathbf{x}, \mathbf{y}, \mathbf{u})$ (13 differential states) $\mathbf{0} = \mathbf{g}(\mathbf{x}, \mathbf{y}, \mathbf{u})$ (8 algebraic variables)

LCL filter + Grid Thevenin $\rightarrow$ PI-VSG OCL (APCL + RPCL) $\rightarrow$ CC-ICL (PI current controller)

1. State variables $\mathbf{x}$ (13)

i_{td}, i_{tq}	VSC-side inductor currents, dq	eq. 1
v_{md}, v_{mq}	Capacitor voltage, dq	eq. 2a
i_{sd}, i_{sq}	Grid-side currents, dq	eqs. 2b+4
ψ	Phase angle difference	eq. 11a
ξ_p	Integral of active power error	eq. 11b
ξ_q	Integral of reactive power error	eq. 9b
i_{sd}^*, i_{sq}^*	LPF current references, dq	eq. 18
ξ_{id}, ξ_{iq}	Current error integrals, dq	eq. 19b

2. Algebraic variables $\mathbf{y}$ (8)

v_{td}, v_{tq}	VSC terminal voltage (CC-ICL output)	eq. 19a
ω_{vsg}	VSG angular speed (PI-VSG)	eq. 11a
E	Virtual EMF amplitude (RPCL)	eq. 9a
v_{sd}, v_{sq}	PCC voltage (grid Thevenin + filter)	eqs. 2b+4
p, q	Active/reactive power at POI	POI

3. External inputs $\mathbf{u}$

p^*, q^*	Active/reactive power references
v_{gd}, v_{gq}	Grid Thevenin voltage, dq (Thevenin source)

4. Parameters (default values, Table I)

L_t	1.25 mH	H	VSC-side inductor	R_t	0.04	Ω	VSC-side resistance
C_f	4 μ F	F	Filter capacitor	R_d	10.0	Ω	Parallel damping
L_s	1.25 mH	H	Grid-side inductor	R_s	0.04	Ω	Grid-side resistance
L_g	2.0 mH	H	Grid Thevenin inductor	R_g	2.0	Ω	Grid Thevenin resistance
k_{pp}	0.0012	rad/s/W	PI-VSG proportional gain	k_{ip}	0.0016	rad/s ² /W	PI-VSG integral gain
k_{pq}	0.0016	V/W	RPCL proportional gain	k_{iq}	0.0163	V/W/s	RPCL integral gain
E_0	230.0	V	Rated virtual EMF	ω_0	100 π	rad/s	Rated angular speed (50 Hz)
k_{pi}^{cc}	1.25	V/A	CC-ICL proportional gain	k_{ii}^{cc}	40.0	V/A	CC-ICL integral gain
G_v	0.0	S	Virtual conductance	B_v	1.25	S	Virtual susceptance
τ_{lpf}	1.6 ms	s	LPF time constant				

A. LCL Filter (eqs. 1, 2a, 2b+4) — 6 states

VSC-side inductor (L_t, R_t):

$$\dot{i}_{td} = (v_{td} - R_t i_{td} + \omega_{vsg} L_t i_{tq} - v_{md}) / L_t$$

$$\dot{i}_{tq} = (v_{tq} - R_t i_{tq} - \omega_{vsg} L_t i_{td} - v_{mq}) / L_t$$

Capacitor (C_f, R_d parallel):

$$\dot{v}_{md} = (i_{td} - i_{sd} - v_{md}/R_d + \omega_{vsg} C_f v_{mq}) / C_f$$

$$\dot{v}_{mq} = (i_{tq} - i_{sq} - v_{mq}/R_d - \omega_{vsg} C_f v_{md}) / C_f$$

Grid-side inductor + grid Thevenin combined (L_s+L_g, R_s+R_g):

$$\dot{i}_{sd} = (v_{md} - v_{gd} - (R_s+R_g) i_{sd} + \omega_{vsg} (L_s+L_g) i_{sq}) / (L_s+L_g)$$

$$\dot{i}_{sq} = (v_{mq} - v_{gq} - (R_s+R_g) i_{sq} - \omega_{vsg} (L_s+L_g) i_{sd}) / (L_s+L_g)$$

B. PI-VSG Outer Control Loop (eqs. 9, 11) — 3 states

APCL — PI controller for active power (eq. 11):

$$\dot{\psi} = -k_{pp}(p^* - p) - k_{ip}\xi_p$$

$$\dot{\xi}_p = p^* - p$$

RPCL — PI controller for reactive power (eq. 9b):

$$\dot{\xi}_q = q^* - q$$

C. CC-ICL — PI Current Controller (eqs. 17-19) — 4 states

Virtual admittance (eq. 17, $e^T = [0, E]^T$, $Y_v = [[G_v, -B_v], [B_v, G_v]] \rightarrow$ substituted inline):

$$\dot{i}_{sd}^* = (-G_v v_{sd} - B_v(E - v_{sq}) - i_{sd}^*) / \tau_{lpf}$$

$$\dot{i}_{sq}^* = (-B_v v_{sd} + G_v(E - v_{sq}) - i_{sq}^*) / \tau_{lpf}$$

PI integrators (eq. 19b):

$$\dot{\xi}_{id} = i_{sd}^* - i_{sd}$$

$$\dot{\xi}_{iq} = i_{sq}^* - i_{sq}$$

D. Algebraic equations $0 = g(x, y, u)$ — 8 equations

CC-ICL terminal voltage (eq. 19a, $L_f = L_t + L_s$):

$$0 = v_{td} - [k_{pi}^{cc}(i_{sd}^* - i_{sd}) + k_{ii}^{cc}\xi_{id} + \omega_{vsg} L_f i_{sq} + v_{sd}]$$

$$0 = v_{tq} - [k_{pi}^{cc}(i_{sq}^* - i_{sq}) + k_{ii}^{cc}\xi_{iq} - \omega_{vsg} L_f i_{sd} + v_{sq}]$$

PI-VSG speed and EMF (from $d\psi/dt = \omega_0 - \omega_{vsg}$, eq. 9a):

$$0 = \omega_{vsg} - \omega_0 - k_{pp}(p^* - p) - k_{ip}\xi_p$$

$$0 = E - E_0 - k_{pq}(q^* - q) - k_{iq}\xi_q$$

PCC voltage (eqs. 2b+4 combined, $\alpha = L_g/(L_s+L_g)$, $R_{mix} = (R_g \cdot L_s - R_s \cdot L_g)/(L_s+L_g)$):

$$0 = v_{sd} - [(1 - \alpha)v_{gd} + \alpha v_{md} + R_{mix} i_{sd}]$$

$$0 = v_{sq} - [(1 - \alpha)v_{gq} + \alpha v_{mq} + R_{mix} i_{sq}]$$

Active and reactive power at POI:

$$0 = p - (v_{sd} i_{sd} + v_{sq} i_{sq})$$

$$0 = q - (v_{sq} i_{sd} - v_{sd} i_{sq})$$
